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Tiresias - An Open-Source Hearing Aid Development Board

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ABSTRACT

Hearing loss is a global public health issue due to its high prevalence and significant negative impacts on quality of life, including social well-being and cognitive health. Despite their crucial role in auditory rehabilitation, hearing aids remain inaccessible to many due to their high costs, particularly in low- and middle-income countries. Existing open-source hearing aid solutions typically rely on bulky, high-power platforms unsuited to practical, wearable applications, creating a significant gap in available technology. This work introduces Tiresias, a compact, low-power, open-source hearing aid development board designed for real-time audio processing using affordable electronics. Tiresias integrates essential hearing aid functionalities—including multichannel compression, digital filtering, adaptive beamforming, Bluetooth connectivity, and physiological monitoring — onto a compact six-layer printed circuit board (PCB). By leveraging widely available hardware components and firmware built upon Nordic's nRF Connect SDK and the Zephyr real-time operating system (RTOS), Tiresias fosters modularity, accessibility, and ease of customization. By addressing existing technological and accessibility challenges, this work provides an effective foundation for open-source hearing aid product innovation, facilitating future research, algorithm refinement, and practical deployment in real-world environments.

1 Introduction

Hearing loss is a global public health issue, affecting over 1.5 billion people worldwide, a number expected to rise to 2.5 billion by 2050. About 80% of people with disabling hearing loss live in low- and middle-income countries (LMICs) [1]. Approximately 26.1%

of the Brazilian population has some hearing impairment, with 6.8% (~14 million people) classified as disabling, thus requiring immediate intervention. The incidence is higher for the elderly, males and individuals with lower socioeconomic status [2]. Sensorineural hearing loss, the most prevalent type, leads to reduced auditory sensitivity, decreased dynamic range,

and impaired frequency selectivity, significantly affecting speech comprehension, particularly in noisy environments. Hearing aids aim to mitigate these effects by applying compression, amplification and integrating directional microphones, noise reduction algorithms, and wireless connectivity.

Brazil's public healthcare system provides hearing aids; however, financial constraints hinder its ability to meet the growing demand. Additionally, the dominance of proprietary technologies restricts accessibility and limits research innovation. Hearing aids play a crucial role in auditory rehabilitation by integrating advanced signal processing techniques for speech enhancement, noise reduction, and wireless connectivity. Despite their benefits, the global hearing aid market is mainly shared by five major companies, whose state-of-the-art devices, developed through extensive institutional knowledge and research, remain costly. This high cost reduces accessibility and contributes to low adoption rates [3].

Proprietary development ecosystems have emerged, yet they pose intellectual property barriers and high adaptation costs for novel algorithms. As a result, promising research frequently struggles to transition into real-world applications. Recognizing this, the National Institute on Deafness and other Communication Disorders (NIDCD) held a workshop advocating for an open, extensible research platform with real-time processing and accessible source code [4]. Such an approach could facilitate the integration of both open and proprietary algorithms, allowing advancements in research, development, and commercial translation.

Furthermore, the widespread availability and growing affordability of high-performance, general-purpose microprocessors and Digital Signal Processors (DSP)s in the mid-2010s were key enablers for deploying and field-testing novel algorithms. In response, the National Institutes of Health (NIH) allocated funding in 2016 to support six different research initiatives aimed at addressing these challenges by developing novel signal processing algorithms employing computing power likely to be available in future generations of consumer devices such as hearing aids and personal sound processors [5, 6]. Several important projects have since emerged, such as Tympan, OpenMHA, UT Dallas Project and Open Speech Platform. Apart from that funding, other initiatives were also introduced.

While some focused on frameworks for software algorithm development, most hardware implementations relied on single-board computers or high-power processors, incompatible with the constraints of hearing aids. Conversely, the few constrained hardware platforms lacked essential functionalities and integration.

To bridge this gap, we propose Tiresias — a low-power, resource-constrained open-source development board for hearing aid prototyping. Tiresias integrates an easy-to-program audio codec supporting both standard hearing aid functions — such as nonlinear amplitude compression and multichannel filtering — and advanced processing techniques, including adaptive filtering and beamforming for noise reduction and speech enhancement.

2 Related Work

Hearing aids are highly specialized embedded systems that must deliver advanced audio processing and user customization while operating under stringent power, size, and latency constraints. Historically, achieving this level of optimization required custom-designed microelectronics. Major manufacturers, leveraging extensive institutional knowledge, either operated as microelectronics design houses or collaborated closely with them to develop proprietary hardware. This reliance on closed, proprietary technology has limited access to flexible, open platforms for prototyping and testing novel algorithms.

By the mid-2000s, embedded electronics saw significant advancements, particularly with the rise of Cortex-M ARM architectures, which combined high performance with low power consumption. Concurrently, Bluetooth 4.0 introduced Bluetooth Low-Energy (BLE), fostering the development of wearables and sensor-equipped wireless devices. Additionally, by the mid-2010s, MEMS microphones had become widely adopted, gradually replacing costlier miniature electret alternatives [7]. These technological advances created an opportunity for assistive audio research to break away from its dependency on proprietary industry solutions.

Despite these innovations, the NIDCD workshop highlighted a persistent gap in hearing aid research: the lack of accessible hardware for real-time audio testing. While software tools and libraries for real-time signal processing were available, the lack of dedicated

prototyping platforms forced researchers to either run algorithms in high-level programming environments on general-purpose computers—limiting real-world applicability—or to bypass crucial optimization and verification steps needed for embedded implementation.

One of the funded initiatives addressing this gap was the UT Dallas Project [8], which focused on developing adaptive signal processing algorithms to enhance speech quality and intelligibility. Their research emphasized Direction of Arrival, Speech Enhancement and wireless communication with smartphones for remote processing. However, it lacked real-time wearable processing, a limitation that contrasts with the open-source Master Hearing Aid (openMHA) [9]. OpenMHA provides a modular framework for hearing aid audio processing, featuring a command-line interface, a dedicated prescription user interface, and various signal processing strategies, including dynamic range compression, noise reduction, and beamforming.

OpenMHA also introduced two hardware implementations: Cape4all [10], a multichannel sound interface designed as an expansion board for the BeagleBone Black single-board computer, and the BatAndCat Corporation's Portable Hearing Laboratory [11], which integrates custom earpieces with hearing aid transducers connected to a single-board computer. While OpenMHA enables real-time audio processing, its reliance on command-line execution and portable computing hardware running Linux makes it impractical for integration into hearing aid compatible hardware.

Similar to OpenMHA, the Open Speech Platform (OSP) provides a real-time hearing aid research suite implemented in C. It includes a hardware platform built around a Snapdragon mobile processor housed in a main unit, which interfaces with external earpieces and an auxiliary ExG¹ sensing unit [12]. However, the bulkiness of this setup imposes limitations on its practicality as a wearable device.

Among NIH-supported projects, Tympan stands out as the only fully wearable open-source hearing aid platform. It features a custom database, framework, smartphone application, and in-ear probe, with hardware built around a Teensy 4.1 directly soldered onto a custom Printed Circuit Board (PCB) and programmed via

¹ExG is a general term referring to electrophysiological measurements such as Electroencephalography, Electrocardiography, and Electromyography. Here, the authors also include Ballistocardiography and Photoplethysmography within this category.

the Arduino IDE. Despite strong performance in gain control, frequency response, and distortion [13], its architecture presents limitations. The platform includes an older Bluetooth-enabled Nordic nRF chip that operates only as a peripheral, requiring audio data to be processed by the Teensy before transmission—adding unnecessary complexity. Furthermore, the NXP processor's high power consumption and 10x10mm footprint reduce its suitability for compact, energy-efficient hearing aid designs.

Many independent work focus on specific functionalities rather than comprehensive hearing aid systems. For instance, [14] presents a hardware design based on the Ambiq Apollo 2 Blue Bluetooth 5.0 microcontroller, aimed at assisting individuals who stutter through Speech Choral techniques. Similarly, [15] introduces a prototype for ExG monitoring, integrating Electrocardiogram electrodes, a triaxial accelerometer, and a LED/photo-detector for Photoplethysmography. While these designs successfully address their objectives, they lack classical hearing aid signal processing, limiting their relevance to broader assistive audio research.

In conclusion, the landscape of hearing aid research has advanced with embedded electronics, wireless communication, and open-source frameworks. While independent projects address specific wearable and hearing challenges, and NIH platforms support algorithm development, neither prioritizes fully open, practical hearing aid hardware. As a result, researchers and emerging market innovators still face a critical gap: the absence of a resource-efficient, accessible, and powerful open hardware platform tailored to the unique demands of hearing aid development and deployment.

3 System Architecture

Individuals with sensorineural hearing loss experience varying listening difficulties, according to the severity of their impairment, often accompanied by reduced dynamic range and frequency selectivity. Over time, hearing aids have evolved from simple active equalizers to sophisticated, non-linear multichannel devices with wireless connectivity and adaptive settings. Multichannel wide dynamic range compression is broadly recognized as an effective strategy for restoring the audibility of soft sounds while preventing discomfort from louder inputs, thus contributing to speech intelligibility in quiet environments, provided that the number

of channels is not excessive [16] and that the system's static and dynamic parameters are appropriately tuned to the user's auditory profile and listening needs.

Beyond classical signal processing, modern hearing aids incorporate multi-microphone techniques, ranging from directional microphones to adaptive beamforming. Although directional microphones are a well-established strategy for improving speech perception in noisy environments, there are situations in which omnidirectional settings are more suitable [17], highlighting the advantages of adaptive approaches. Additionally, beamforming has proven useful for noise reduction algorithms [18].

With the advent of Industry 4.0 and the Internet of Things (IoT), a variety of radio-enabled microcontrollers have emerged. Among them, Bluetooth devices have become widely available; however, Low Energy Audio (LE Audio) was only introduced in Bluetooth 5.2. This standard has gained traction among audio developers due to its Isochronous Channels, which enable lower latency and improved synchronization, independent multi-stream audio for binaural streaming and Auracast™ systems, and the Low Complexity Communications Codec (LC3), which provides better audio quality at lower power consumption.

Wireless communication enhances both binaural signal processing—leveraging interaural cues via Bluetooth—and interactions with external devices like smartphones. Studies such as [19] suggests that inter-device connectivity can enhance directionality and noise suppression. Remote processing has also been explored, though maintaining low latency remains a challenge. Nevertheless, outsourcing non-real-time functions, such as user configurations, has been successfully demonstrated [20]. Furthermore, the integration of physiological sensors for ExG monitoring continues to be an emerging trend in hearing aid technology.

Although many modern microcontrollers include Floating-Point Units (FPU) for audio processing, specialized Integrated Circuits (ICs) designed for audio—such as audio codecs—offer mixed-signal capabilities. These ICs integrate analog filters, amplifiers, and high-resolution converters, often accompanied by dedicated DSP cores. Compared to general-purpose microcontrollers, audio codecs typically feature higher

bit-depth and sampling rates while also providing pre-defined audio processing blocks, simplifying development and integration.

Multi-microphone approaches can be achieved with a good balance of quality, cost, and power efficiency through MEMS microphones. Additionally, a novel type of audio sensor has emerged—the Voice Pick-Up (VPU). These vibration sensors capture speech directly from the user's skin through bone conduction, providing speech data that can support voice reconstruction or enhance speech detection from the user [12].

Furthermore, Inertial Measurement Units (IMUs) play an increasingly important role in hearing aids. They can track head movements and body posture, providing valuable data for adaptive audio processing, such as automatic beamforming adjustments. Moreover, IMUs contribute to ExG data sensing, either independently or in conjunction with specialized electrode-based sensors. Beyond hearing enhancement, IMUs also support traditional health monitoring applications, including fall detection and gait analysis, which are particularly relevant for elderly users.

Historically, most hearing aids relied on coin cell batteries. However, modern hearing aids and earbuds predominantly employ rechargeable Li-Po batteries. These batteries require Power Management Integrated Circuits (PMICs), which regulate voltage levels, manage power distribution among the device's components, and control recharging when connected to an external power source.

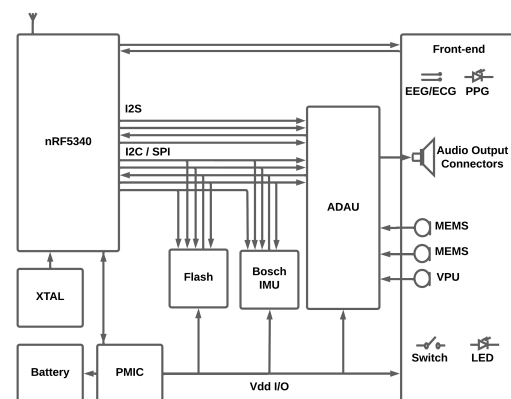


Fig. 1: Tiresias PCB system architecture.

Considering the outlined specifications and constraints, the system design was based on the integration of a Bluetooth-enabled System-on-Chip (SoC) and a fixed-DSP-function audio codec, which handles most of the audio processing. This architecture balances development flexibility and ease of implementation with the power efficiency and extended battery life required for wearable devices. In accordance with these design principles, the Tiresias PCB integrates:

- A Nordic nRF5340: Dual-core Bluetooth 5.4 System-on-Chip (SoC) with FPU support;
- An Analog Devices ADAU1787: Audio codec featuring multiple input and output channels and two independent, programmable DSP cores with pre-defined processing functions;
- Two PUI-Audio MEMS microphones;
- A Knowles V2S200D VPU sensor;
- A Bosch BMI270 IMU for motion tracking and sensing;
- A Macronix Microelectronics ultra low power 16Mb Flash memory;
- A Nordic nPM1100 PMIC;
- Extensive connectors and test points to facilitate prototyping.

Figure 1 illustrates the system architecture in a block diagram. The nRF microprocessor acts as the central controller, interfacing with all subsystems. It manages the PMIC, which regulates and distributes battery power across the PCB. A Flash memory module is included to enable over-the-air firmware updates, accommodating the application's memory footprint. Communication with peripherals—including the Flash, IMU, and audio codec—is handled via standard protocols such as I²C and SPI. Digital audio data is exchanged with the codec, that integrates transducers and communicates using an Inter-IC Sound (I²S) interface. Additionally, user interface components (e.g., switches and LEDs) and optional external sensors are grouped in a front-end block.

Tiresias is built using Nordic Semiconductor's nRF Connect Software Development Kit (NCS), which integrates the Zephyr Real-Time Operating System. This

SDK provides a comprehensive suite of tools, including libraries, third-party device drivers, and graphical interfaces to streamline development. A key advantage of Zephyr's modularity is its ability to simplify driver implementation and enhance RTOS functionalities such as thread management and task scheduling through its dedicated libraries.

Analog Devices also offers an SDK for its audio processors—SigmaStudio, an open-access, block diagram-based programming suite designed to interface with the company's proprietary DSP cores. This SDK features an intuitive GUI that enables users to interconnect fixed-function audio processors, facilitating the creation of custom algorithms. It also includes signal sources, switches, and simulation probes, allowing for real-time analysis, such as plotting the frequency response of an implemented filter.

The ADAU1787 audio codec, selected for this project, supports both standard hearing aid processing blocks, such as compressors and digital filters, and advanced techniques like beamforming and adaptive filtering for applications such as noise and feedback canceling. It features two dedicated memory banks: Program RAM, which stores operational codes executed by the core, and Parameter RAM, which holds adjustable parameters such as filter coefficients and volume settings. Using SigmaStudio, designers can dynamically modify parameters in real-time, enabling immediate auditory feedback—for instance, adjusting a compressor's rate and instantly perceiving its effect.

SigmaStudio can also export the generated program and parameters as C header files, facilitating the programming of the codec by the SoC. Using nRF Connect, the authors developed a custom driver for Inter-Integrated Circuit (I²C) communication between the nRF5340 SoC and the ADAU1787 audio codec [21], allowing the SoC to load the content of these header files into the codec's Program and Parameter RAM. This enables users to design a data flow through an intuitive graphical interface and seamlessly deploy it onto a custom PCB.

The system is designed on a 25x40 mm PCB, a size comparable to 3.7V 650mAh battery packs, facilitating chassis integration. The design was carried out in Altium Designer, and a 3D render of the PCB is shown in Figure 2. All of the design files are publicly available at [22]. The board features two independent audio outputs for external balanced armature receivers,

LC3-streamed audio, this system will classify acoustic scenes and correspondingly adjust Parameter RAM settings dynamically, which enables scene-specific configurations for optimized intelligibility and auditory comfort. Future scalability efforts will include mobile app integration and cloud-based dashboards, enhancing user control and overall experience.

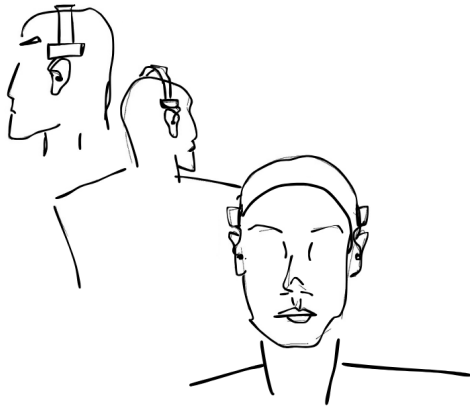


Fig. 3: A suggested wearable case for the prototype.

5 Conclusions

This paper presents Tiresias, an open-source, resource-constrained hardware platform designed for wearable audio processing, with applications in hearing aid research and product development. By integrating state-of-the-art, off-the-shelf components, Tiresias provides a flexible and accessible tool for researchers and innovators in the field of hearing solutions.

By combining a Bluetooth-enabled microcontroller and a powerful yet easy-to-program DSP-enabled audio codec, the platform offloads standard hearing aid signal processing tasks, such as compression, filtering, and beamforming, from the microcontroller, reducing both development complexity and power consumption. The platform also integrates onboard IMU and VPU sensors, along with a range of connectors and test points, enabling basic physiological signal acquisition for future integration of ExG-based applications and expanding its functional scope.

Tiresias addresses the gap in development tools for hearing aid-compatible hardware by offering a modular, extensible, and open-source solution. Its compatibility

with widely available components democratizes access to advanced hearing aid technologies, accelerating the path from research to final product development.

Following validation, future enhancements will include audiologically validated processing schemes, a custom wearable chassis, expanded connectivity options, and user-friendly applications for improved accessibility and usability.

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